

Derek Lankeaux

Data Scientist | Applied Statistics | Healthcare AI, LLM Evaluation & Applied ML

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SUMMARY

Applied Statistics M.S. candidate focused on applying Python, SQL, experimental design, and machine learning to healthcare-relevant and operational data problems. Built four independent technical case studies spanning diagnostic-ML benchmarking, LLM evaluation, retrieval-augmented generation, and AI-safety evaluation. Strong at framing questions, building reproducible evaluation workflows, quantifying uncertainty, and communicating results and limitations to technical and non-technical stakeholders.

TECHNICAL SKILLS

Data and statistics: Python, R, SQL, Pandas, Polars, NumPy, SciPy, statsmodels; experimental design, power analysis, hypothesis testing, bootstrap confidence intervals, inter-rater reliability, Bayesian hierarchical modeling, MCMC diagnostics, calibration

Machine learning: scikit-learn, XGBoost, LightGBM, AdaBoost, Optuna, SMOTE, SHAP; feature engineering, model selection, cross-validation, threshold analysis, explainability

LLM, APIs, and systems: OpenAI, Anthropic, Hugging Face, LangChain, FastAPI, MLflow, Qdrant, BM25, ColBERT, Docker, Kubernetes, Git/GitHub; LLM evaluation, retrieval-augmented generation, API-based development, retrieval quality, observability design, cost/latency trade-offs

EDUCATION

M.S., Applied Statistics — Rochester Institute of Technology, expected 2026 Coursework: Bayesian methods, machine learning, experimental design, deep learning, statistical learning theory, and computational statistics.

SELECTED TECHNICAL PROJECTS

AI Safety Red-Team Evaluation

Independent research case study · April 2026

- Designed a two-stage LLM evaluation workflow that combines ensemble labels with supervised harm classification across 12,500 response pairs and six harm categories.
- Reported $\alpha = 0.81$ inter-rater reliability and 96.8% held-out classification accuracy; separated agreement, model performance, and uncertainty analysis in the evaluation.
- Used Bayesian hierarchical modeling and feature attribution to support risk review and audit-oriented reporting.

Methods: LLM evaluation, XGBoost/stacking, PyMC, SHAP, MLflow

LLM Ensemble Textbook Bias Detection

Independent research case study · April 2026

- Evaluated 4,500 textbook passages with a rubric-based LLM ensemble, generating 67,500 ratings for reliability and uncertainty analysis.
- Reported Krippendorff's $\alpha = 0.84$ and modeled publisher-level effects with Bayesian partial pooling and MCMC diagnostics.
- Documented an expert-review-oriented workflow for interpreting disagreement and high-uncertainty passages.

Methods: LLM-as-judge, PyMC, ArviZ, FastAPI, MLflow

Breast Cancer Classification Benchmark

Independent research case study · April 2026

- Benchmarked eight ensemble classifiers on the Wisconsin Diagnostic Breast Cancer dataset using preprocessing, feature selection, calibration, and cross-validation.
- Reported 99.12% held-out accuracy and 0.9987 ROC-AUC for the best configuration, together with calibration and threshold analyses.
- Framed the work as a diagnostic decision-support benchmark; it is not a clinical device or patient-validation study.

Methods: scikit-learn, XGBoost, LightGBM, AdaBoost, Optuna, SMOTE, SHAP

RAG Production Pipeline

Independent systems-design case study · April 2026

- Designed a hybrid RAG architecture using dense retrieval, BM25, re-ranking, grounding checks, and confidence calibration.
- Reported 96.3% Recall@10 and 94.2% citation precision in the documented evaluation, with latency, throughput, drift, and failure-mode analysis.
- Specified observability, privacy, and rollback considerations needed before a production deployment.

Methods: Qdrant, BM25, ColBERT, OpenAI, Kafka, Kubernetes, Prometheus, MLflow

PUBLICATIONS

Technical report	Version	Date
AI Safety Red-Team Evaluation	2.0.0	April 2026
LLM Ensemble Textbook Bias Detection	4.0.0	April 2026
Breast Cancer Classification	4.0.0	April 2026
RAG Production Pipeline	3.0.0	April 2026

Availability: Remote/hybrid · Seeking 2026 opportunities · Authorized to work in the United States